Basics of FE-Analysis
BK10A6400
Lecture 5: 2D solid (plane) elements (Cook's book chapter 3, Hakala p. 260)

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Sign convention

- Right-hand rule. Nodal displacements.
- Interpretation of member forces.
- Sign rule in statics and strength of material.

ANSYS ↔ FEMAP / NX Nastran element equivalence

Type / Application	**ANSYS**	**FEMAP / NX Nastran**
Rod (axial only)	LINK180	CROD
Beam (Timoshenko)	BEAM188 / BEAM189	CBEAM / CBAR
2D solid (linear)	PLANE42 (old)	CQUAD4 / CTRIA3
2D solid (linear)	PLANE182	CQUAD4 / CTRIA3
2D solid (quadratic)	PLANE183	CQUAD8 / CTRIA6
2D sxisymmetric	PLANE182 / 183 (KEYOPT=3)	CAX4 / CAX8
Shell / Plate	SHELL181	CQUAD4 / CQUAD8 (Shell)
3D solid	SOLID185	CHEXA
3D solid	SOLID186	CPENTA
3D solid	SOLID187	CTETRA

CROD for axial only, CBEAM/CBAR for full 3D beam, plane stress/strain via Property (FEMAP) or KEYOPT (ANSYS).

Example: A simple cantilever problem solved with one 4 DOF beam element: completed together in the lecture

RECAP: The elemental stiffness matrix for a 2D Euler-Bernoulli 4 DOF beam element:

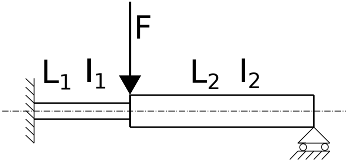
$$\overline{\boldsymbol{K}}_{EB4DOF} = \begin{pmatrix} \frac{12EI}{L^3} & \frac{6EI}{L^2} & -\frac{12EI}{L^3} & \frac{6EI}{L^2} \\ \frac{6EI}{L^2} & \frac{4EI}{L} & -\frac{6EI}{L^2} & \frac{2EI}{L} \\ -\frac{12EI}{L^3} & -\frac{6EI}{L^2} & \frac{12EI}{L^3} & -\frac{6EI}{L^2} \\ \frac{6EI}{L^2} & \frac{2EI}{L} & -\frac{6EI}{L^2} & \frac{4EI}{L} \end{pmatrix}$$

Example: A simple cantilever problem . . .

Example: A simple cantilever problem . . .

Example: Beam structure with two beam elements

The aim is to solve nodal displacements and member forces using a two-dimensional **two-nodded Euler-Bernoulli beam element (6 DOF)** in MATLAB. For simplicity, a mesh of two beam elements is only used to analyze the beam structure shown in the figure.



- Discretized by two 6 DOF's E-B elements
- 3 nodes, 3 DOF's at node, $3 \times 3 = 9$ DOF's
- 4 fixed/constrained nodal displacements.

Elemental stiffness matrices

Element 1:

$$
\overline{\boldsymbol{K}}^{(1)} = \begin{array}{c} \begin{array}{cccccc} u_1 & v_1 & \theta_1 & u_2 & v_2 & \theta_2 \end{array} \\ \begin{pmatrix} \frac{E_1 A_1}{L_1} & 0 & 0 & -\frac{E_1 A_1}{L_1} & 0 & 0 \\ 0 & \frac{12 E_1 I_1}{L_1^3} & \frac{6 E_1 I_1}{L_1^2} & 0 & -\frac{12 E_1 I_1}{L_1^3} & \frac{6 E_1 I_1}{L_1^2} \\ 0 & \frac{6 E_1 I_1}{L_1^2} & \frac{4 E_1 I_1}{L_1} & 0 & -\frac{6 E_1 I_1}{L_1^2} & \frac{2 E_1 I_1}{L_1} \\ -\frac{E_1 A_1}{L_1} & 0 & 0 & \frac{E_1 A_1}{L_1} & 0 & 0 \\ 0 & -\frac{12 E_1 I_1}{L_1^3} & -\frac{6 E_1 I_1}{L_1^2} & 0 & \frac{12 E_1 I_1}{L_1^3} & -\frac{6 E_1 I_1}{L_1^2} \\ 0 & \frac{6 E_1 I_1}{L_1^2} & \frac{2 E_1 I_1}{L_1} & 0 & -\frac{6 E_1 I_1}{L_1^2} & \frac{4 E_1 I_1}{L_1} \end{pmatrix} \end{array} \begin{array}{c} \\ u_1 \\ v_2 \\ \theta_1 \\ u_2 \\ v_2 \\ \theta_2 \end{array}
$$

Elemental stiffness matrices

Element 2:

$$
\overline{\boldsymbol{K}}^{(2)} = \begin{array}{c c}
\begin{array}{cccccc} u_2 & v_2 & \theta_2 & u_3 & v_3 & \theta_3 \end{array} & \\
\begin{pmatrix}
\frac{E_2 A_2}{L_2} & 0 & 0 & -\frac{E_2 A_2}{L_2} & 0 & 0 \\
0 & \frac{12 E_2 I_2}{L_2^3} & \frac{6 E_2 I_2}{L_2^2} & 0 & -\frac{12 E_2 I_2}{L_2^3} & \frac{6 E_2 I_2}{L_2^2} \\
0 & \frac{6 E_2 I_2}{L_2^2} & \frac{4 E_2 I_2}{L_2} & 0 & -\frac{6 E_2 I_2}{L_2^2} & \frac{2 E_2 I_2}{L_2} \\
-\frac{E_2 A_2}{L_2} & 0 & 0 & \frac{E_2 A_2}{L_2} & 0 & 0 \\
0 & -\frac{12 E_2 I_2}{L_2^3} & -\frac{6 E_2 I_2}{L_2^2} & 0 & \frac{12 E_2 I_2}{L_2^3} & -\frac{6 E_2 I_2}{L_2^2} \\
0 & \frac{6 E_2 I_2}{L_2^2} & \frac{2 E_2 I_2}{L_2} & 0 & -\frac{6 E_2 I_2}{L_2^2} & \frac{4 E_2 I_2}{L_2}
\end{pmatrix} &
\begin{array}{c} u_2 \\ v_2 \\ \theta_2 \\ u_3 \\ v_3 \\ \theta_3 \end{array}
\end{array}
$$

Transformation

The transformation from a local coordinate system to a global coordinate system. Now $\theta = 0$ so the transformation is not necessarily needed because $\boldsymbol{T}$ is a unit matrix then.

$$\boldsymbol{K}_G^{(1)} = \boldsymbol{T}^T(0)\overline{\boldsymbol{K}}^{(1)}\boldsymbol{T}(0) \quad ; \quad \boldsymbol{K}_G^{(2)} = \boldsymbol{T}^T(0)\overline{\boldsymbol{K}}^{(2)}\boldsymbol{T}(0) \qquad (1)$$

Assembling

$$
\boldsymbol{K}_G = \begin{array}{c}
\begin{array}{ccccccccc} u_1 & v_1 & \theta_1 & u_2 & v_2 & \theta_2 & u_3 & v_3 & \theta_3 \end{array} \\
\left(\begin{array}{ccccccccc}
k_{11}^{(1)} & k_{12}^{(1)} & k_{13}^{(1)} & k_{14}^{(1)} & k_{15}^{(1)} & k_{16}^{(1)} & 0 & 0 & 0 \\
k_{21}^{(1)} & k_{22}^{(1)} & k_{23}^{(1)} & k_{24}^{(1)} & k_{25}^{(1)} & k_{26}^{(1)} & 0 & 0 & 0 \\
k_{31}^{(1)} & k_{32}^{(1)} & k_{33}^{(1)} & k_{34}^{(1)} & k_{35}^{(1)} & k_{36}^{(1)} & 0 & 0 & 0 \\
k_{41}^{(1)} & k_{42}^{(1)} & k_{43}^{(1)} & k_{44}^{(1)} + k_{11}^{(2)} & k_{45}^{(1)} + k_{12}^{(2)} & k_{46}^{(1)} + k_{13}^{(2)} & k_{14}^{(2)} & k_{15}^{(2)} & k_{16}^{(2)} \\
k_{51}^{(1)} & k_{52}^{(1)} & k_{53}^{(1)} & k_{54}^{(1)} + k_{21}^{(2)} & k_{55}^{(1)} + k_{22}^{(2)} & k_{56}^{(1)} + k_{23}^{(2)} & k_{24}^{(2)} & k_{25}^{(2)} & k_{26}^{(2)} \\
k_{61}^{(1)} & k_{62}^{(1)} & k_{63}^{(1)} & k_{64}^{(1)} + k_{31}^{(2)} & k_{65}^{(1)} + k_{32}^{(2)} & k_{66}^{(1)} + k_{33}^{(2)} & k_{34}^{(2)} & k_{35}^{(2)} & k_{36}^{(2)} \\
0 & 0 & 0 & k_{41}^{(2)} & k_{42}^{(2)} & k_{43}^{(2)} & k_{44}^{(2)} & k_{45}^{(2)} & k_{46}^{(2)} \\
0 & 0 & 0 & k_{51}^{(2)} & k_{52}^{(2)} & k_{53}^{(2)} & k_{54}^{(2)} & k_{55}^{(2)} & k_{56}^{(2)} \\
0 & 0 & 0 & k_{61}^{(2)} & k_{62}^{(2)} & k_{63}^{(2)} & k_{64}^{(2)} & k_{65}^{(2)} & k_{66}^{(2)}
\end{array}\right)
\end{array}
\begin{array}{c} \\ u_1 \\ v_1 \\ \theta_1 \\ u_2 \\ v_2 \\ \theta_2 \\ u_3 \\ v_3 \\ \theta_3 \end{array}
$$

Force vector

$$\boldsymbol{f}_G = \begin{array}{c c} \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ -F \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} & \begin{array}{c} u_1 \\ v_1 \\ \theta_1 \\ u_2 \\ v_2 \\ \theta_2 \\ u_3 \\ v_3 \\ \theta_3 \end{array} \end{array}$$

Elimination of linear constraints

Fixed nodal displacements: $u_1 = v_1 = \theta_1 = v_3 = 0$. Let's eliminate corresponding rows and columns regarding FE-matrices.

$$\boldsymbol{f}_{G_{red}} = \begin{array}{cc} \begin{pmatrix} 0 \\ -F \\ 0 \\ 0 \\ 0 \end{pmatrix} & \begin{array}{c} u_2 \\ v_2 \\ \theta_2 \\ u_3 \\ \theta_3 \end{array} \end{array}$$

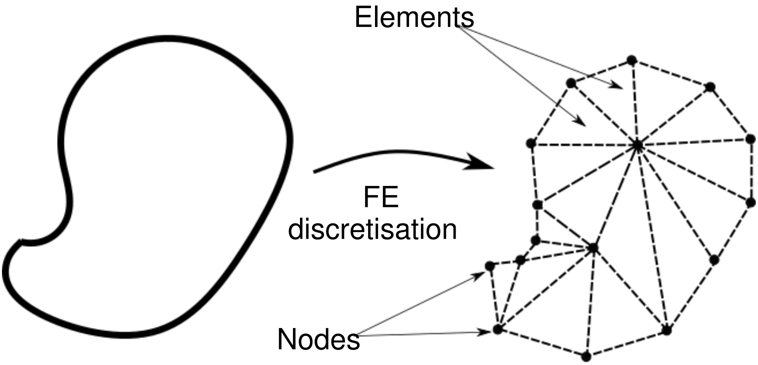
Assembling

$$
\boldsymbol{K}_{G_{red}} = \begin{array}{cc}
\begin{array}{ccccc} \quad u_2 \quad & \quad v_2 \quad & \quad \theta_2 \quad & u_3 & \theta_3 \end{array} & \\
\begin{pmatrix}
k_{44}^{(1)} + k_{11}^{(2)} & k_{45}^{(1)} + k_{12}^{(2)} & k_{46}^{(1)} + k_{13}^{(2)} & k_{14}^{(2)} & k_{16}^{(2)} \\
k_{54}^{(1)} + k_{21}^{(2)} & k_{55}^{(1)} + k_{22}^{(2)} & k_{56}^{(1)} + k_{23}^{(2)} & k_{24}^{(2)} & k_{26}^{(2)} \\
k_{64}^{(1)} + k_{31}^{(2)} & k_{65}^{(1)} + k_{32}^{(2)} & k_{66}^{(1)} + k_{33}^{(2)} & k_{34}^{(2)} & k_{36}^{(2)} \\
k_{41}^{(2)} & k_{42}^{(2)} & k_{43}^{(2)} & k_{44}^{(2)} & k_{46}^{(2)} \\
k_{61}^{(2)} & k_{62}^{(2)} & k_{63}^{(2)} & k_{64}^{(2)} & k_{66}^{(2)}
\end{pmatrix} &
\begin{array}{c} u_2 \\ v_2 \\ \theta_2 \\ u_3 \\ \theta_3 \end{array}
\end{array}
$$

Note that $\boldsymbol{K}_{G_{red}}$ is symmetric i.e. $k_{ij} = k_{ji}$ (if not, there is a typo) and $\det \boldsymbol{K}_{G_{red}} > 0$ (if not, check BC's).
And then we solve displacements (in a global coordinate system).

$$
\boldsymbol{u}_{G_{red}} = \boldsymbol{K}_{G_{red}}^{-1} \boldsymbol{f}_{G_{red}}
$$

Schematic FE discretisation of an arbitrary 2D structure

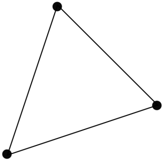


Some two-dimensional solid (plane) elements

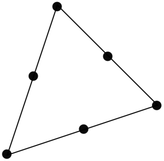
- Constant strain triangle (CST), 3 nodes, 6 DOFs
- Linear strain triangle (LST), 6 nodes, 12 DOFs
- Bilinear quadrilateral (Q4), 4 nodes, 8 DOFs
- 8-node biquadratic quadrilateral (Q8), 8 nodes, 16 DOFs
- 9-node biquadratic quadrilateral (Q9), 9 nodes, 18 DOFs
- Bilinear quadrilateral (improved by "bubble modes"), 4 nodes, 8 DOFs (+ 2 internal DOFs that are eliminated by static condensation).

Some two-dimensional solid (plane) elements

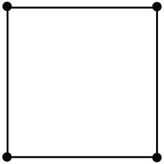
CST



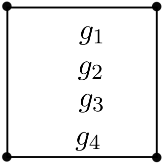
LST



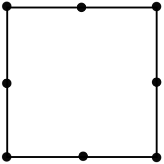
Q4



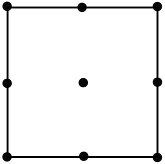
Q6



Q8



Q9



Notations for strains and stresses in a vector form for a 2D case:

$$\boldsymbol{\varepsilon} = \begin{pmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ 2\varepsilon_{12} \end{pmatrix} = \begin{pmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \gamma_{xy} \end{pmatrix}$$

$$\boldsymbol{\sigma} = \begin{pmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{12} \end{pmatrix} = \begin{pmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \tau_{xy} \end{pmatrix}$$

Strains in 2D

Linear relationship (small displacements and rotations) between strains and displacements:

$$\varepsilon_{xx} = \frac{\partial u_h}{\partial x} \quad ; \quad \varepsilon_{yy} = \frac{\partial v_h}{\partial y} \quad ; \quad \gamma_{xy} = \frac{\partial u_h}{\partial y} + \frac{\partial v_h}{\partial x}$$

Strain visualization in 2D: to be completed in the lecture

Strains in 2D

- Small angle assumption: $\tan \alpha_1 = \frac{\partial u_h}{\partial y} \approx \alpha_1$, $\tan \alpha_2 = \frac{\partial v_h}{\partial x} \approx \alpha_2$

Strains and stresses in 2D

A linear strain-displacement relationship can be written in a matrix form as

$$\underbrace{\begin{pmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \gamma_{xy} \end{pmatrix}}_{\boldsymbol{\varepsilon}} = \underbrace{\begin{pmatrix} \frac{\partial}{\partial x} & 0 \\ 0 & \frac{\partial}{\partial y} \\ \frac{\partial}{\partial y} & \frac{\partial}{\partial x} \end{pmatrix}}_{\boldsymbol{\partial}} \underbrace{\begin{pmatrix} u_h \\ v_h \end{pmatrix}}_{\boldsymbol{u}_h} \tag{2}$$

We remember that

$$\boldsymbol{u}_h = \boldsymbol{N}\boldsymbol{u} \quad ; \quad \boldsymbol{\varepsilon} = \underbrace{\boldsymbol{\partial}\boldsymbol{N}}_{\boldsymbol{B}}\boldsymbol{u} \tag{3}$$

where $\boldsymbol{B}$ is so-called the strain-displacement matrix that can be written as

$$\boldsymbol{B} = \begin{pmatrix} \frac{\partial N_1}{\partial x} & 0 & \frac{\partial N_2}{\partial x} & 0 & \frac{\partial N_3}{\partial x} & 0 & \dots \\ 0 & \frac{\partial N_1}{\partial y} & 0 & \frac{\partial N_2}{\partial y} & 0 & \frac{\partial N_3}{\partial y} & \dots \\ \frac{\partial N_1}{\partial y} & \frac{\partial N_1}{\partial x} & \frac{\partial N_2}{\partial y} & \frac{\partial N_2}{\partial x} & \frac{\partial N_3}{\partial y} & \frac{\partial N_3}{\partial x} & \dots \end{pmatrix} \tag{4}$$

Displacement field

Displacement field $\boldsymbol{u}_h = \{u_h, v_h\}^T$ is interpolated as follows

$$\begin{cases} u_h(x,y) := N_1 u_1 + N_2 u_2 + \ldots + N_n u_n \\ v_h(x,y) := N_1 v_1 + N_2 v_2 + \ldots + N_n v_n \end{cases} \tag{5}$$

and same in a matrix form:

$$\underbrace{\begin{pmatrix} u_h \\ v_h \end{pmatrix}}_{\boldsymbol{u}_h} = \underbrace{\begin{pmatrix} N_1 & 0 & N_2 & 0 & \ldots & N_n u_n \\ 0 & N_1 & 0 & N_2 & \ldots & N_n v_n \end{pmatrix}}_{\boldsymbol{N}} \underbrace{\begin{pmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \\ \vdots \\ u_n \\ v_n \end{pmatrix}}_{\boldsymbol{u}} \tag{6}$$

$$\boldsymbol{u}_h = \boldsymbol{N}\boldsymbol{u} \tag{7}$$

Linear elasticity

Linear relationship between stresses and strains.

$$\boldsymbol{\sigma} = \boldsymbol{D}\boldsymbol{\varepsilon} \tag{8}$$

Plane stress

It is assumed that a plane is thin (in z-direction) and constant thickness. Then there is no stress through the plate's thickness. In the case of plane stress i.e. $\sigma_{zz} = 0, \sigma_{xz} = 0, \sigma_{yz} = 0$,the elasticity matrix $\boldsymbol{D}$ for a linear elastic and isotropic material is written as:

$$\boldsymbol{D} = \frac{E}{1-\nu^2}\begin{pmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1-\nu}{2} \end{pmatrix} \tag{9}$$

Plain strain

Plane strain assumptions provided good representations because the in-plane strains are developed from full linear (Hooke's law) 3D elasticity.

- The out-of-plane (z-direction) strains are set to zero (analogy to an object that is constrained by rigid walls in z-direction
- $\varepsilon_{zz} = \varepsilon_{xz} = \varepsilon_{yz} = 0$

$$\boldsymbol{D} = \frac{E}{(1+\nu)(1-\nu)} \begin{pmatrix} 1-\nu & \nu & 0 \\ \nu & 1-\nu & 0 \\ 0 & 0 & \frac{1-2\nu}{2} \end{pmatrix} \tag{10}$$

Strain energy and elemental stiffness matrix

$$W_{int} = \frac{1}{2}\int_V \boldsymbol{\sigma}\boldsymbol{\varepsilon}\,\mathrm{d}V = \frac{1}{2}\boldsymbol{u}^T \underbrace{\int_V \boldsymbol{B}^T\boldsymbol{D}\boldsymbol{B}\mathrm{d}V}_{\overline{\boldsymbol{K}}}\,\boldsymbol{u} \tag{11}$$

The stiffness matrix can be found via the strain-displacement matrix:

$$\overline{\boldsymbol{K}} = \int_V \boldsymbol{B}^T\boldsymbol{D}\boldsymbol{B}\,\mathrm{d}V \tag{12}$$

or more generally by using:

$$\boldsymbol{f}_{int} = \frac{\partial W_{int}}{\partial \boldsymbol{u}} \quad ; \quad \overline{\boldsymbol{K}} = \frac{\partial \boldsymbol{f}_{int}}{\partial \boldsymbol{u}}$$

Note that the stiffness matrix is a product of integration over volume (area for 2D solids).

Bilinear four node element

- Four nodes, 2 DOF's at node, 8 DOF's
- Linear interpolation for u_h, v_h
- Cannot describe pure bending - overly stiff in bending (spurious shear deformation)
- Suffers from shear locking

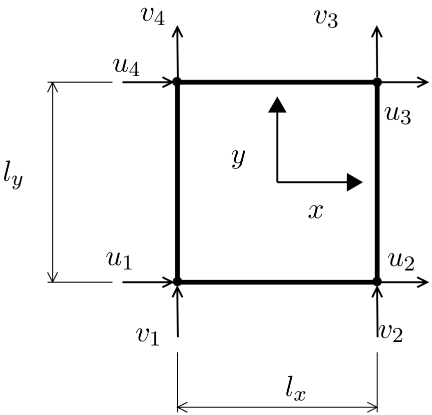
Basis functions for a four-node bilinear element (Q4):

$$\boldsymbol{p} = \{1, x, y, xy\} \tag{13}$$

Displacement field interpolations:

$$\begin{cases} u_h(x,y) := a_0 + a_1 x + a_2 y + a_3 xy \\ v_h(x,y) := b_0 + b_1 x + b_2 y + b_3 xy \end{cases} . \tag{14}$$

Bilinear four node element



Example: shape functions for a four-node bilinear plane element

Basis functions for a four-node bilinear element (Q4):

$$\boldsymbol{p} = \{1, x, y, xy\} \tag{15}$$

Displacement field interpolations:

$$\begin{cases} u_h(x,y) := a_0 + a_1 x + a_2 y + a_3 xy \\ v_h(x,y) := b_0 + b_1 x + b_2 y + b_3 xy \end{cases}.$$

Displacement interpolation is same for the both displacements $a \Rightarrow$ same shape functions. Four unknowns, four eqs:

$$\begin{cases} u_h(-l_x/2, -l_y/2) := a_0 - \frac{1}{2} a_1 l_x - \frac{1}{2} a_2 l_y + \frac{1}{4}\, a_3\, l_x\, l_y = u_1 \\ u_h(l_x/2, -l_y/2) := a_0 + \frac{1}{2} a_1 l_x - \frac{1}{2} a_2 l_y - \frac{1}{4}\, a_3\, l_x\, l_y = u_2 \\ u_h(l_x/2, l_y/2) := a_0 + \frac{1}{2} a_1 l_x + \frac{1}{2} a_2 l_y + \frac{1}{4}\, a_3\, l_x\, l_y = u_3 \\ u_h(-l_x/2, l_y/2) := a_0 - \frac{1}{2} a_1 l_x + \frac{1}{2} a_2 l_y - \frac{1}{4}\, a_3\, l_x\, l_y = u_4 \end{cases}.$$

Example: shape functions for a four-node bilinear plane element

$$\begin{cases} u_h(-l_x/2, -l_y/2) := a_0 - \frac{1}{2}a_1 l_x - \frac{1}{2}a_2 l_y + \frac{1}{4}\, a_3\, l_x\, l_y = u_1 \\ u_h(l_x/2, -l_y/2) := a_0 + \frac{1}{2}a_1 l_x - \frac{1}{2}a_2 l_y - \frac{1}{4}\, a_3\, l_x\, l_y = u_2 \\ u_h(l_x/2, l_y/2) := a_0 + \frac{1}{2}a_1 l_x + \frac{1}{2}a_2 l_y + \frac{1}{4}\, a_3\, l_x\, l_y = u_3 \\ u_h(-l_x/2, l_y/2) := a_0 - \frac{1}{2}a_1 l_x + \frac{1}{2}a_2 l_y - \frac{1}{4}\, a_3\, l_x\, l_y = u_4 \end{cases} .$$

In a matrix form:

$$\underbrace{\begin{pmatrix} 1 & -1/2\, l_x & -1/2\, l_y & 1/4\, l_x\, l_y \\ 1 & 1/2\, l_x & -1/2\, l_y & -1/4\, l_x\, l_y \\ 1 & 1/2\, l_x & 1/2\, l_y & 1/4\, l_x\, l_y \\ 1 & -1/2\, l_x & 1/2\, l_y & -1/4\, l_x\, l_y \end{pmatrix}}_{\boldsymbol{A}} \underbrace{\begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{pmatrix}}_{\boldsymbol{a}} = \underbrace{\begin{pmatrix} u_1 \\ u_1 \\ u_3 \\ u_4 \end{pmatrix}}_{\boldsymbol{u}_u} . \tag{16}$$

Example: shape functions for a four-node bilinear plane element

By solving a_i:s:

$$\boldsymbol{a} = \boldsymbol{A}^{-1}\boldsymbol{u}_u \tag{17}$$

and substituting them into the Eq.(18):

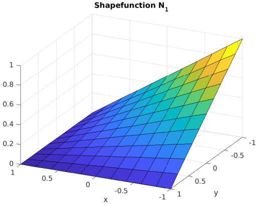
$$u_h = \boldsymbol{a}^T\boldsymbol{p} = \boldsymbol{A}^{-T}\boldsymbol{u}_u^T\boldsymbol{p}^T = \underbrace{\boldsymbol{A}^{-T}\boldsymbol{p}}_{\boldsymbol{N}}\boldsymbol{u}_u \tag{18}$$

where shape functions $\boldsymbol{N}$ can be written as

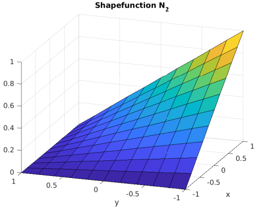
$$N_1 = \frac{(a-x)(b-y)}{4ab} \quad ; N_2 = \frac{(a+x)(b-y)}{4ab} \quad ; N_3 = \frac{(a+x)(b+y)}{4ab} \quad ; N_4 = \frac{(a-x)(b+y)}{4ab}$$

where $a = l_x/2$ and $b = l_y/2$.

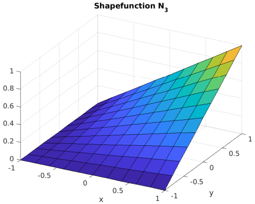
Example: shape functions for a four-node bilinear plane element



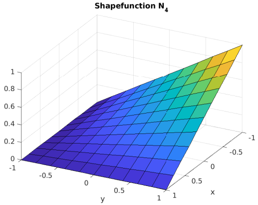
(a) $N_1(-1,-1) = 1$



(b) $N_2(1,-1) = 1$



(c) $N_3(1,1) = 1$



(d) $N_4(-1,1) = 1$

Strain energy

Displacement field $\boldsymbol{u}_h$ is interpolated as follows
Displacement field:

$$\underbrace{\begin{pmatrix} u_h \\ v_h \end{pmatrix}}_{\boldsymbol{u}_h} = \underbrace{\begin{pmatrix} N_1 & 0 & N_2 & 0 & N_3 & 0 & N_4 & 0 \\ 0 & N_1 & 0 & N_2 & 0 & N_3 & 0 & N_4 \end{pmatrix}}_{\boldsymbol{N}_m} \underbrace{\begin{pmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \\ u_3 \\ v_3 \\ u_4 \\ v_4 \end{pmatrix}}_{\boldsymbol{u}} \tag{19}$$

$$\boldsymbol{u}_h = \boldsymbol{N}_m \boldsymbol{u} \tag{20}$$

Strain energy and elemental stiffness matrix

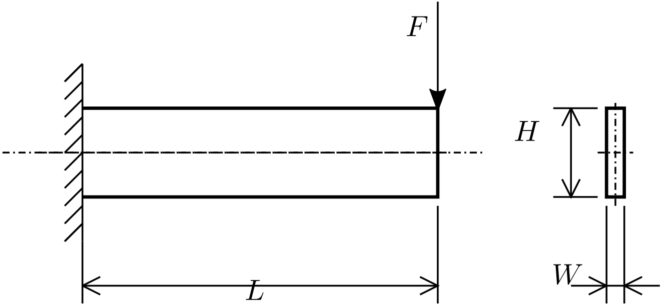
$$W_{int} = \frac{1}{2}\int_V \boldsymbol{\sigma}\boldsymbol{\varepsilon}\,\mathrm{d}V = \frac{1}{2}\boldsymbol{u}^T \underbrace{l_z \int_x \int_y \boldsymbol{B}^T \boldsymbol{D} \boldsymbol{B}\,\mathrm{d}y\mathrm{d}x}_{\overline{\boldsymbol{K}}}\,\boldsymbol{u} \tag{21}$$

$$\overline{\boldsymbol{K}} = \int_V \boldsymbol{B}^T \boldsymbol{D} \boldsymbol{B}\,\mathrm{d}V = l_z \int_x \int_y \boldsymbol{B}^T \boldsymbol{D} \boldsymbol{B}\,\mathrm{d}y\mathrm{d}x = l_z \int_{-l_x}^{l_x} \int_{-l_y}^{l_y} \boldsymbol{B}^T \boldsymbol{D} \boldsymbol{B}\,\mathrm{d}y\mathrm{d}x \tag{22}$$

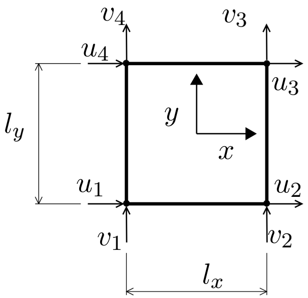
or more generally as:

$$\boldsymbol{f}_{int} = \frac{\partial W_{int}}{\partial \boldsymbol{u}} \quad ; \quad \overline{\boldsymbol{K}} = \frac{\partial \boldsymbol{f}_{int}}{\partial \boldsymbol{u}}$$

Example: Cantilever beam analyzed by Q4



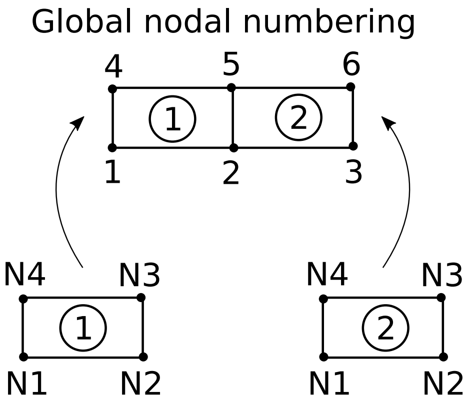
Bilinear four node element



$$\underbrace{\begin{pmatrix} u_h \\ v_h \end{pmatrix}}_{\boldsymbol{u}_h} = \underbrace{\begin{pmatrix} N_1 & 0 & N_2 & 0 & N_3 & 0 & N_4 & 0 \\ 0 & N_1 & 0 & N_2 & 0 & N_3 & 0 & N_4 \end{pmatrix}}_{\boldsymbol{N}} \underbrace{\begin{pmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \\ u_3 \\ v_3 \\ u_4 \\ v_4 \end{pmatrix}}_{\boldsymbol{u}} \tag{23}$$

$$\boldsymbol{u}_h = \boldsymbol{N}\boldsymbol{u} \tag{24}$$

Example: Cantilever beam analyzed by Q4



Element 1 consists of nodes 1,2,5,4:

$$
\begin{array}{cc}
 & \begin{array}{cccccccc} u_1 & v_1 & u_2 & v_2 & u_5 & v_5 & u_4 & v_4 \end{array} \\
\overline{\boldsymbol{K}}^{(1)} = & \begin{pmatrix}
k_{11}^{(1)} & k_{12}^{(1)} & k_{13}^{(1)} & k_{14}^{(1)} & k_{15}^{(1)} & k_{16}^{(1)} & k_{17}^{(1)} & k_{18}^{(1)} \\
k_{21}^{(1)} & k_{22}^{(1)} & k_{23}^{(1)} & k_{24}^{(1)} & k_{25}^{(1)} & k_{26}^{(1)} & k_{27}^{(1)} & k_{28}^{(1)} \\
k_{31}^{(1)} & k_{32}^{(1)} & k_{33}^{(1)} & k_{34}^{(1)} & k_{35}^{(1)} & k_{36}^{(1)} & k_{37}^{(1)} & k_{38}^{(1)} \\
k_{41}^{(1)} & k_{42}^{(1)} & k_{43}^{(1)} & k_{44}^{(1)} & k_{45}^{(1)} & k_{46}^{(1)} & k_{47}^{(1)} & k_{48}^{(1)} \\
k_{51}^{(1)} & k_{52}^{(1)} & k_{53}^{(1)} & k_{54}^{(1)} & k_{55}^{(1)} & k_{56}^{(1)} & k_{57}^{(1)} & k_{58}^{(1)} \\
k_{61}^{(1)} & k_{62}^{(1)} & k_{63}^{(1)} & k_{64}^{(1)} & k_{65}^{(1)} & k_{66}^{(1)} & k_{67}^{(1)} & k_{68}^{(1)} \\
k_{71}^{(1)} & k_{72}^{(1)} & k_{73}^{(1)} & k_{74}^{(1)} & k_{75}^{(1)} & k_{76}^{(1)} & k_{77}^{(1)} & k_{78}^{(1)} \\
k_{81}^{(1)} & k_{82}^{(1)} & k_{83}^{(1)} & k_{84}^{(1)} & k_{85}^{(1)} & k_{86}^{(1)} & k_{87}^{(1)} & k_{88}^{(1)}
\end{pmatrix}
\begin{array}{c} u_1 \\ v_1 \\ u_2 \\ v_2 \\ u_5 \\ v_5 \\ u_4 \\ v_4 \end{array}
\end{array}
$$

Element 2 consists of nodes 2,3,6,5:

$$
\overline{\boldsymbol{K}}^{(2)} = \begin{array}{cc}
\begin{array}{cccccccc} u_2 & v_2 & u_3 & v_3 & u_6 & v_6 & u_5 & v_5 \end{array} & \\
\begin{pmatrix}
k_{11}^{(2)} & k_{12}^{(2)} & k_{13}^{(2)} & k_{14}^{(2)} & k_{15}^{(2)} & k_{16}^{(2)} & k_{17}^{(2)} & k_{18}^{(2)} \\
k_{21}^{(2)} & k_{22}^{(2)} & k_{23}^{(2)} & k_{24}^{(2)} & k_{25}^{(2)} & k_{26}^{(2)} & k_{27}^{(2)} & k_{28}^{(2)} \\
k_{31}^{(2)} & k_{32}^{(2)} & k_{33}^{(2)} & k_{34}^{(2)} & k_{35}^{(2)} & k_{36}^{(2)} & k_{37}^{(2)} & k_{38}^{(2)} \\
k_{41}^{(2)} & k_{42}^{(2)} & k_{43}^{(2)} & k_{44}^{(2)} & k_{45}^{(2)} & k_{46}^{(2)} & k_{47}^{(2)} & k_{48}^{(2)} \\
k_{51}^{(2)} & k_{52}^{(2)} & k_{53}^{(2)} & k_{54}^{(2)} & k_{55}^{(2)} & k_{56}^{(2)} & k_{57}^{(2)} & k_{58}^{(2)} \\
k_{61}^{(2)} & k_{62}^{(2)} & k_{63}^{(2)} & k_{64}^{(2)} & k_{65}^{(2)} & k_{66}^{(2)} & k_{67}^{(2)} & k_{68}^{(2)} \\
k_{71}^{(2)} & k_{72}^{(2)} & k_{73}^{(2)} & k_{74}^{(2)} & k_{75}^{(2)} & k_{76}^{(2)} & k_{77}^{(2)} & k_{78}^{(2)} \\
k_{81}^{(2)} & k_{82}^{(2)} & k_{83}^{(2)} & k_{84}^{(2)} & k_{85}^{(2)} & k_{86}^{(2)} & k_{87}^{(2)} & k_{88}^{(2)}
\end{pmatrix} &
\begin{array}{c} u_2 \\ v_2 \\ u_3 \\ v_3 \\ u_6 \\ v_6 \\ u_5 \\ v_5 \end{array}
\end{array}
$$

Assembling and BC's

Global stiffness matrix is a size of 12×12. Nodes 2 and 5 are common nodes. Nodal displacements are fixed at nodes 1 and 4.

$$
\boldsymbol{K}_{G_{red}} = \begin{array}{c}
\begin{array}{cccccccc}
u_2 & v_2 & u_3 & v_3 & u_5 & v_5 & u_6 & v_6
\end{array} \\
\begin{pmatrix}
k^{(1)}_{33} + k^{(2)}_{11} & k^{(1)}_{34} + k^{(2)}_{12} & k^{(2)}_{13} & k^{(2)}_{14} & k^{(1)}_{35} + k^{(2)}_{17} & k^{(1)}_{36} + k^{(2)}_{18} & k^{(2)}_{1,5} & k^{(2)}_{16} \\
k^{(1)}_{43} + k^{(2)}_{21} & k^{(1)}_{44} + k^{(2)}_{22} & k^{(2)}_{23} & k^{(2)}_{24} & k^{(1)}_{45} + k^{(2)}_{27} & k^{(1)}_{46} + k^{(2)}_{28} & k^{(2)}_{25} & k^{(2)}_{26} \\
k^{(2)}_{31} & k^{(2)}_{32} & k^{(2)}_{33} & k^{(2)}_{34} & k^{(2)}_{37} & k^{(2)}_{38} & k^{(2)}_{35} & k^{(2)}_{36} \\
k^{(2)}_{41} & k^{(2)}_{42} & k^{(2)}_{43} & k^{(2)}_{44} & k^{(2)}_{47} & k^{(2)}_{48} & k^{(2)}_{45} & k^{(2)}_{46} \\
k^{(1)}_{53} + k^{(2)}_{71} & k^{(1)}_{54} + k^{(2)}_{72} & k^{(2)}_{73} & k^{(2)}_{74} & k^{(1)}_{55} + k^{(2)}_{77} & k^{(1)}_{56} + k^{(2)}_{78} & k^{(2)}_{75} & k^{(2)}_{76} \\
k^{(1)}_{63} + k^{(2)}_{81} & k^{(1)}_{64} + k^{(2)}_{82} & k^{(2)}_{83} & k^{(2)}_{84} & k^{(1)}_{65} + k^{(2)}_{87} & k^{(1)}_{66} + k^{(2)}_{88} & k^{(2)}_{85} & k^{(2)}_{86} \\
k^{(2)}_{51} & k^{(2)}_{52} & k^{(2)}_{53} & k^{(2)}_{54} & k^{(2)}_{57} & k^{(2)}_{58} & k^{(2)}_{55} & k^{(2)}_{56} \\
k^{(2)}_{61} & k^{(2)}_{62} & k^{(2)}_{63} & k^{(2)}_{64} & k^{(2)}_{67} & k^{(2)}_{68} & k^{(2)}_{65} & k^{(2)}_{66}
\end{pmatrix}
\end{array}
\begin{array}{c}
\\ u_2 \\ v_2 \\ u_3 \\ v_3 \\ u_5 \\ v_5 \\ u_6 \\ v_6
\end{array}
$$

Force vector and solution

After taking account for boundary conditions $u_1 = v_1 = u_4 = v_4 = 0$, global force vector can be written as:

$$\boldsymbol{f}_{G_{red}} = \begin{array}{c} \\ \\ \\ \\ \\ \\ \\ \\ \end{array} \begin{pmatrix} 0 \\ 0 \\ 0 \\ -F/2 \\ 0 \\ 0 \\ 0 \\ -F/2 \end{pmatrix} \begin{array}{c} u_2 \\ v_2 \\ u_3 \\ v_3 \\ u_5 \\ v_5 \\ u_6 \\ v_6 \end{array}$$

Analytical solution

$$\frac{d^2v}{dx^2} = -\frac{M(x)}{EI(x)} - \frac{\kappa_{xy}q(x)}{GA} \qquad (25)$$

Maximum deflection for the cantilever beam under applied force at the free end can be found as:

$$\delta_{max} = \underbrace{\frac{FL^3}{3EI}}_{\text{bending}} + \kappa_{xy}\underbrace{\frac{FL}{GA}}_{\text{shear}} \qquad (26)$$

where κ_{xy} is a shear correction factor (needed to use in the beams based on the first-order shear deformation theory (FSDT)). It can be approximated to be $6/5$ for a rectangular cross-section.